

Activity-4

CO-2 ASSESSMENT TOOL-1

25/06/2026

① Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

- What is the mean of the data? What is the median?
- What is the mode of the data?
comment on the data's modality (i.e., bimodal, trimodal, etc)
- What is the midrange of the data?
- Can you find (roughly) the first quartile (Q_1) and the third quartile (Q_3) of the data?
- Give the five-number summary of the data.
- Show a boxplot of the data.

a) mean $\Rightarrow \frac{1}{n} \sum_{i=1}^n x_i$

809
27

$$13 + 15 + 16 + 16 + 19 + 20 + 20 + 21 + 22 +$$

$$22 + 25 + 25 + 25 + 25 + 30 + 33 + 33 + 35 +$$

$$35 + 35 + 35 + 36 + 40 + 45 + 46 + 52 + 70$$

$$= \frac{1}{n} \sum_{i=1}^{27} x_i$$

$$= \frac{1}{27} \times 809$$

$$= 29.96$$

mean = 30

$$\text{median} = n + \frac{1}{2}$$

$$= 27 + \frac{1}{2}$$

$$= 14$$

median = 25

b) mode =

25 and 35 are coming repeatedly, as only two values are repeating they are called as bimodal

max + min

2

$$\text{midrange} = \frac{70 + 13}{2}$$

$$\text{midrange} = 41.5$$

$$Q_1 = 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25$$

$$\begin{aligned} Q_1 = \text{median} &= n + \frac{1}{2} \\ &= 13 + \frac{1}{2} \\ &= 7 \end{aligned}$$

$$Q_1 = 20$$

$$Q_3 = 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70$$

$$\begin{aligned} Q_3 = \text{median} &= n + \frac{1}{2} \\ &= 13 + \frac{1}{2} \\ &= 7 \end{aligned}$$

$$Q_3 = 35$$

e) Five-number summary:

min a_1 , median a_3 , max

13, 20, 25, 35, 70

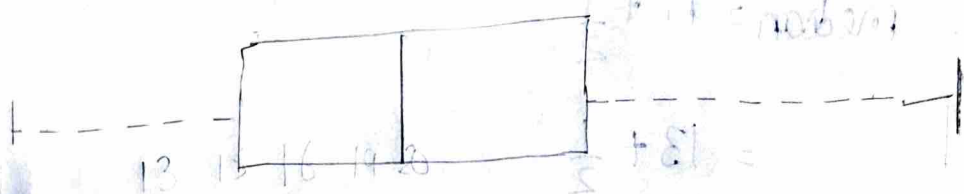
f) Boxplot:

$$a_1 = 13$$

$$a_2 = 20$$

$$\text{median} = 25$$

$$\text{min} = 13, \text{max} = 70$$



13 15 16 19 20 21 22 25 30 33 35 36 40 45 46 52 70

$$a_5 = 70$$

$$\frac{1}{2} + n$$

$$\frac{1}{2} + 13$$

$$F =$$



Given the following measurements for the variable age:

18, 22, 25, 42, 28, 43, 33, 35, 56, 28,

standardize the variable by the following:

Compute the mean absolute deviation of age

$$\text{mean} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$18 + 22 + 25 + 42 + 28 + 43 + 33 + 35 + 56 + 28 = 330$$

$$= \frac{1}{10} \times 330$$

$$\text{mean} = 33$$

x (age)	$ x - \bar{x} $
18	$ 18 - 33 = 15$
22	$ 22 - 33 = 11$
25	$ 25 - 33 = 8$
42	$ 42 - 33 = 9$
28	$ 28 - 33 = 5$
43	$ 43 - 33 = 10$
33	$ 33 - 33 = 0$
35	$ 35 - 33 = 2$
56	$ 56 - 33 = 23$
28	$ 28 - 33 = 5$
	<u>88</u>

Mean Absolute Deviation

$$= \frac{\sum |x - \bar{x}|}{n} = \frac{88}{10} = 8.8$$

mean absolute deviation = 8.8

$$0.88 = 8.8 + 8.8 + 8.8 + 8.8 + 8.8 + 8.8 + 8.8 + 8.8 + 8.8 + 8.8$$



$$0.88 \times \frac{1}{10}$$

$$8.8$$

	$ x - \bar{x} $	$(x - \bar{x})^2$
12	$ 88 - 81 $	81
11	$ 88 - 82 $	55
8	$ 88 - 85 $	25
8	$ 88 - 84 $	54
2	$ 88 - 85 $	85
01	$ 88 - 84 $	44
0	$ 88 - 88 $	88
5	$ 88 - 88 $	88
85	$ 88 - 82 $	22
8	$ 88 - 86 $	88
88		

③ Suppose that the data for analysis includes the attributes age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70. Use smoothing by bin means to smooth the above data, using a bin depth of 3. Illustrate your steps.

$$\text{mean} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$= \frac{1}{27} \times 809$$

$$= 29.96$$

$$\text{mean} = 30$$

Total number of values = 27

Bin depth = 3

$$\text{count of bin} = \frac{\text{Occurrence}}{\text{bin depth}}$$

$$= \frac{27}{3}$$

$$\text{count of bin} = 9$$

Bin	Value	
B1	13, 15, 16	$13+15+16/3 = 14.67$
B2	16, 19, 20	$16+19+20/3 = 18.33$
B3	20, 21, 22	$20+21+22/3 = 21.00$
B4	22, 25, 25	$22+25+25/3 = 24.00$
B5	25, 25, 30, 33	$25+25+30/3 = 26.66$
B6	33, 33, 35	$33+33+35/3 = 33.66$
B7	35, 35, 35	$35+35+35/3 = 35.00$
B8	36, 40, 45	$36+40+45/3 = 40.33$
B9	46, 52, 70	$46+52+70/3 = 56.00$

Smoothed Value - Replace the mean all the ~~tree~~ values in the bin

14.67, 14.67, 14.67

18.33, 18.33, 18.33

21.00, 21.00, 21.00

24.00, 24.00, 24.00

26.66, 26.66, 26.66

33.66, 33.66, 33.66

35.00, 35.00, 35.00

40.33, 40.33, 40.33

56.00, 56.00, 56.00

④ Use the two methods below to normalize the following group of data:

200, 300, 400, 600, 1000 (a) min-max

normalization by setting min=0 and max=1

(b) Z-score normalization

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

$$x' = \frac{x - 200}{800}$$

original value	Calculations
200	$200 - 200 / 800 = 0$
300	$(300 - 200) / 800 = 0.125$
400	$(400 - 200) / 800 = 0.25$
600	$(600 - 200) / 800 = 0.5$
1000	$(1000 - 200) / 800 = 1$

$$Z = \frac{x - \mu}{\sigma}$$

mean

σ

$$\mu = \frac{200 + 300 + 400 + 600 + 1000}{5} = 500$$

5

x'	$x - \mu$	$(x - \mu)^2$
200	-300	90,000
300	-200	40,000
400	-100	10,000
600	100	10,000
1000	500	2,50,000

$$\sum (x - \mu)^2 = 400000$$

$$\sigma^2 = \frac{400000}{5} = 80000$$

$$\sigma = \sqrt{80000} \approx 282.84$$

Original value	Calculation	Z-Score
200	$(x - \mu) / \sigma = \frac{(200 - 500)}{282.84}$	-1.06
300	$(300 - 500) / 282.84$	-0.70
400	$(400 - 500) / 282.84$	-0.35
600	$(600 - 500) / 282.84$	0.35
1000	$(1000 - 500) / 282.84$	1.77

Z-scores are:

-1.06, -0.70, -0.35, 0.35, 1.77

⑤ Suppose a group of 12 Baby Price records has been sorted as follows:

5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 215
Partition them into three bins by each of the following methods.

- (a) Equal frequency Partitioning
- (b) Equal-width Partitioning
- (c) clustering

Total records = 12

No of bins = 3

a) Equal frequency Partitioning

Number of records = 12

Number of bins = 3

Records per bin = $\frac{12}{3} = 4$

Bin values

B1 5, 10, 11, 13

B2 15, 35, 50, 55

B3 72, 92, 204, 215

b) Equal width partitioning

$$\text{width} = \frac{\text{maximum} - \text{minimum}}{\text{Number of bins}}$$

$$\text{width} = \frac{215 - 5}{3}$$

$$\text{width} = 70$$

The intervals are: First value + 70 - 1

Bin 1: 5 - 74

Bin 2: 75 - 144

Bin 3: 145 - 215

Bin	Interval	values
Bin 1	5 - 74	5, 10, 11, 13, 15, 35, 50, 55, 72
Bin 2	75 - 144	92
Bin 3	145 - 215	204, 215

Cluster (bin)	values
Cluster 1	5, 10, 11, 13, 15
Cluster 2	35, 50, 55, 72, 92
Cluster 3	204, 215

⑥ Suppose a hospital tested the age and body fat data for 18 randomly selected adults with the following result

age 23 23 27 27 39 41 47 49 50

%fat 9.2 26.5 7.8 17.8 31.4 25.9 27.4 27.2 31.2

age 52 54 54 56 57 58 58 60 61

%fat 34.6 42.5 28.8 33.4 30.2 34.1 32.9 41.2 35.7

a) Calculate the mean, median and standard deviation of age and %fat

b) Draw the boxplots for age and %fat

c) Draw a scatter plot and a Q-Q plot based on these two variables

d) Normalize the two variables based on Z-score normalization

e) Calculate the correlation coefficient (Pearson Product moment coefficient). Are these two variables positively or negatively correlated?

Answer
2/1/19

1) 23	9.2
2) 23	86.5
3) 27	7.8
4) 27	17.8
5) 39	31.4
6) 41	25.9
7) 47	27.4
8) 49	27.2
9) 50	31.2
10) 52	34.6
11) 54	42.5
12) 54	28.8
13) 56	33.4
14) 57	30.2
15) 58	34.1
16) 58	32.9
17) 60	41.2
18) 61	35.7

$$\begin{aligned}
 \text{mean of age} &= \frac{1}{n} \sum_{i=1}^n x_i \\
 &= \frac{1}{18} \times 836 \\
 &= 46.44
 \end{aligned}$$

$$\text{mean \% fat} = \frac{1}{18} \times 83.78$$

Median of age

$$= n + \frac{1}{2}$$

$$= 18 + \frac{1}{2}$$

$$= 18.5 \text{ } 30.70 \text{ } 51$$

Median of fat

$$= n + \frac{1}{2}$$

$$= 18 + \frac{1}{2}$$

$$= 18.5 \text{ } 30.70 \text{ } 30.7$$

$$\min = 23$$

$$Q_1 = 33$$

$$\text{Median} = 51 \text{ (Age)}$$

$$Q_3 = 57$$

$$\max = 61$$

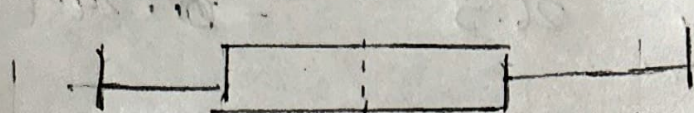
$$\text{Median} = 30.70 \text{ (fat)}$$

Five summary

Min, Q_1 , Med, Q_3 , Max

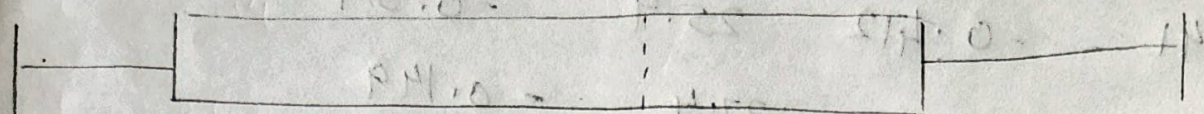
$$23, 33, 51, 57, 61$$

Box plot for bat



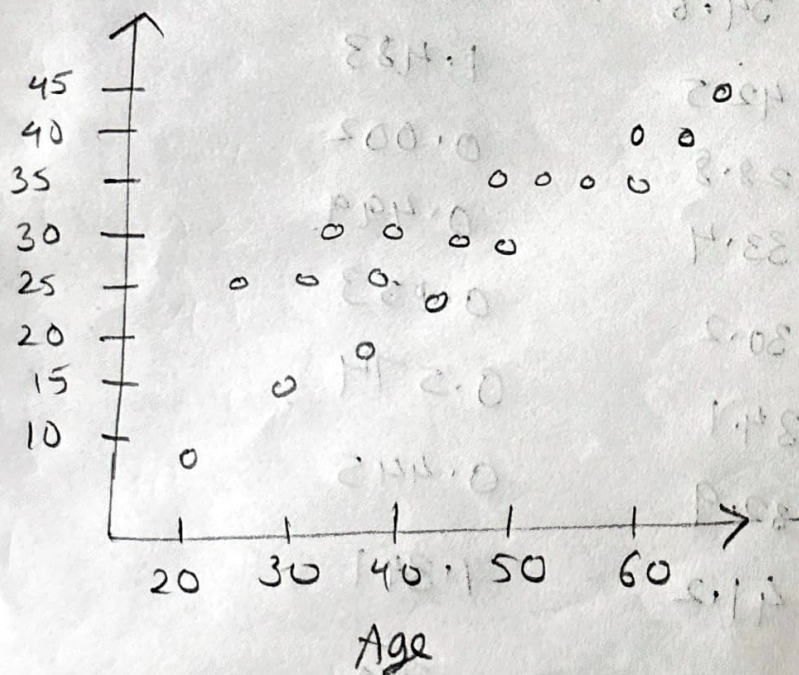
26.3 26.5 30.7 34.7 42.5

Box plot for age



23 23 27 27 39 41 47 49 50 52 54 54 56 57 58 58 60 61

Scatter plot



Z-score normalization

$$Z = \frac{x - \mu}{\sigma}$$

$$= \frac{\text{Age} - 46.444}{13.219}$$

Age	Z (Age)	% fat	2 (1.66)
23	-1.774	9.5	-2.034
23	-1.774	26.5	-0.247
23	-1.471	7.8	-2.268
27	-1.471	17.8	-1.187
27	-0.563	31.4	0.283
39	-0.412	25.9	-0.311
41	0.042	27.4	-0.149
47	0.193	27.2	-0.171
49	0.269	31.2	+0.261
56	0.420	34.6	0.629
52	0.571	42.5	1.483
54	0.571	28.8	0.002
54	0.723	33.4	0.499
56	0.799	30.2	0.153
57	0.874	34.1	0.574
58	0.874	32.9	0.445
59	1.025	41.2	1.341
60	1.101	35.7	0.748
61			

After Normalization

mean = 0 std deviation = 1

11/11/21

11/11/21

e) Pearson Correlation Coefficient

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

$$r = 0.818 > 0$$

The variables have a positive correlation